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LITTORAL PHYSICS TEACHERS' ASSOCIATION (LIPTA)

REGIONAL MOCK EXAMINATION

GENERAL CERTIFICATE OF EDUCATION EXAMINATION

PHYSICS 2
0780

MARCH 2025 **ADVANCED LEVEL**

Subject Title	PHYSICS
Paper No	2
Subject Code	0780

Three hours

Answer ALL questions

Section I is designed to be answered in 1 ½ hour, Section II in 40 minutes and Section III in 50 minutes.

You are advised to divide your time accordingly.

You are reminded of the necessity for good English and orderly presentation in your answers.

In calculations you must show all the steps in your working, giving your answer at each stage.

Non-programmable calculators and GCE Board formulae booklets are allowed.

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SECTION I (1 ½ HOUR)

Answer ALL Questions in this Section.

1) a) Explain why a unit is essential in expressing the value of a physical quantity but cannot uniquely identify the quantity.

b) When an electron accelerated by a high potential difference, V, enters perpendicularly to a uniform magnetic field of flux density B, it is deflected through a circular path of radius, r. The specific charge of an electron is given by;

$$Q = \frac{2V}{B^2r^2}$$

Show that the equation is homogeneous **(6 marks)**

2) a) Explain how rubber differs from steel in terms of its mechanical properties.

b) Figure 1 shows a steel wire of length 100 cm fixed horizontally between two supports A and B, 100 cm apart. When a mass of 8.0 kg is hung from the mid-point of the wire, the middle of the wire sags a distance of 2.0 cm.

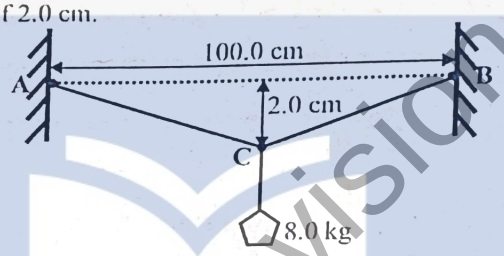


Figure 1

i) Calculate the total extension of the wire.

ii) If the diameter of the wire is 2.8 mm, determine the Young's modulus of the material of the wire. **(6 marks)**

3) a) What is meant by an emission spectrum?

b) Figure 2 shows part of the emission spectrum from a hydrogen gas in a laboratory. The vertical lines indicate the wavelengths in nm.

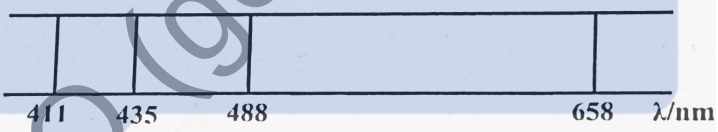


Figure 2

i) Show clearly the region of the electromagnetic spectrum where these lines are found.

ii) Explain how this emission spectrum provides evidence for the existence of discrete energy levels for the electron in the hydrogen atom.

iii) Determine the highest energy of the emission photon. **(6 marks)**

4) a) Define temperature coefficient of resistance of a material.

b) A copper wire has a resistance of 5.0 Ω at ice point and a temperature coefficient of resistance of $3.9 \times 10^{-3} K^{-1}$. It is connected in series with an aluminium wire of resistance of 10.0 Ω at ice point and a temperature coefficient of resistance of $4.3 \times 10^{-3} K^{-1}$.

i) The wires in series are connected to a 24.0 V power supply. Determine the power dissipated in the circuit when the temperature of both wires is increased to 30 ^{0}C.

ii) Sketch on the same axis, graphs, showing the variation of the resistance of copper and the resistance of aluminium with temperature between 0 ^{0}C and 30 ^{0}C. **(6 marks)**

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5) a) State two conditions necessary for an emf to be induced in a conductor in a magnetic field.

b) Figure 3 shows a rectangular coil of cross-sectional area 8.0 cm² and 70 turns placed in a uniform magnetic field of flux density 0.50 T. The normal to the coil makes an angle $\theta = 60°$ to the magnetic field.

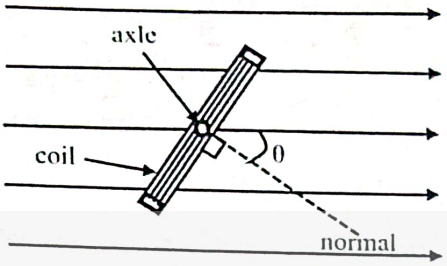


Figure 3

i) Calculate the magnetic flux linkage for the coil.

ii) The coil is rotated at constant speed, causing an emf to be induced. Sketch a graph to show how the induced emf varies with angle θ during one complete rotation of the coil, starting when $\theta = 0^0$.

(6 marks)

ANSWER EITHER 6 a, b and c, OR 6 d, e and f

EITHER 6 a, b and c.

6) a) i) State the second law of thermodynamics in terms of entropy. **(2 marks)**

Figure 4 shows a fixed mass of an ideal gas compressed isothermally from A to B and then the pressure increased from B to C. The temperature at A is 130 K.

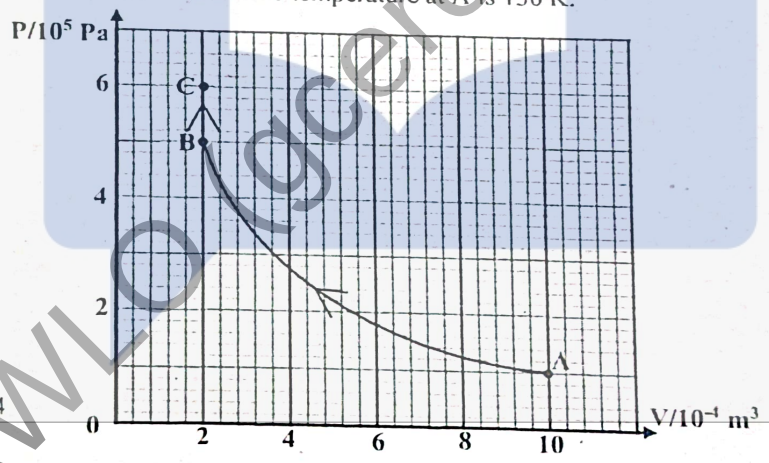


Figure 4

ii) Name the process undergone by the gas from B to C.

iii) Determine the entropy change when the gas is compressed isothermally. **(7 marks)**

b) In an experiment to determine the permittivity of free space (ε_0), a pair of insulated parallel metal plates of area 0.15 m² and spacing 1.2 mm is repeatedly charged and discharged 50 times each second using a reed switch, as shown in figure 4.

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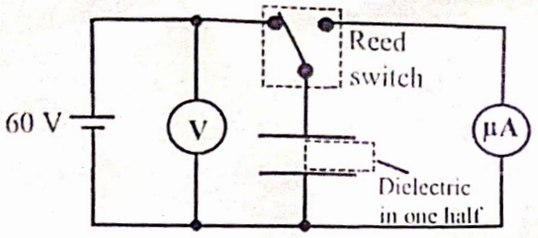


Figure 5

The switch is used to charge the plates up to 60 V and then to discharge the plates through a microammeter. The reading on the microammeter was 3.32 μA.

i) Calculate the capacitance of the plates as charge is stored on the plates each cycle.
ii) Determine the value of ε_0.
iii) The spacing between the plates is completely filled with a dielectric material, causing the current to increase to 10.2 μA. Calculate the relative permittivity of the dielectric.
iv) What would be the microammeter reading if the space between the plates is only half filled with the dielectric material, as shown in figure 5. **(9 marks)**

c) i) State Newton's third law of motion.
ii) A body is falling in the gravitational field of the earth. If the weight of the body is regarded as the 'action', what is the 'reaction' to it in the sense of Newton's third law? **(2 marks)**

OR 6 d, e and f

6) d) i) State Newton's law of universal gravitation. **(2 marks)**

Figure 6 shows the variation of gravitational potential with distance r from the surface of a dense compact planet.

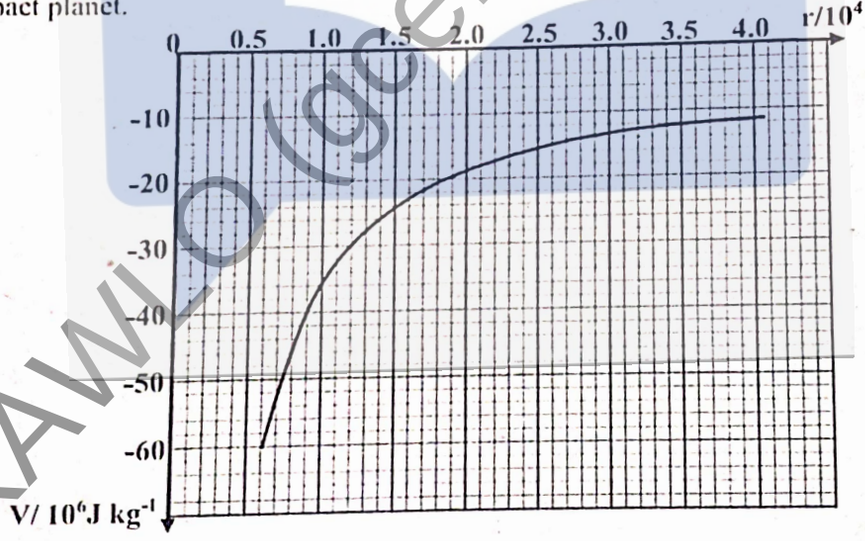


Figure 6

ii) State the radius of the planet.
iii) Determine the force that will act on a mass of 1200 kg placed at a distance of 1.5×10^4 km from the centre of the planet. **(7 marks)**

e) An experiment is setup to measure the wavelength of light in air and in water using a diffraction grating (AB). Figure 7(a), shows the diffraction of a parallel beam monochromatic light incident normally on a plane diffraction grating in air. Figure 7(b), shows the arrangement in water of refractive index 1.33. In water, the angle $\alpha < 30^0$, and the angle $\beta < 30^0$.

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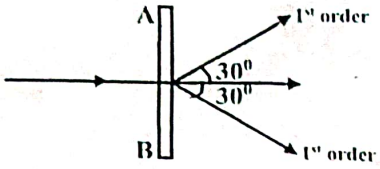


Figure 7a

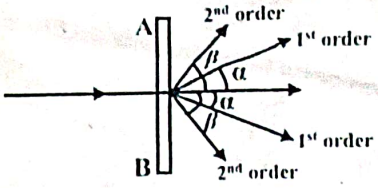


Figure 7b

i) Explain the change ($\alpha < 30^0$ and $\beta < 30^0$) caused by the water.
Given that the grating has 8000 lines per centimetre, calculate;
ii) the wavelength of light in air.
iii) the wavelength of light in water.
iv) the angles α and β. **(9 marks)**

f) i) Define binding energy of a nucleus.
ii) What is the implication of a low binding energy per nucleon on the stability of a nucleus? **(2 marks)**

SECTION II (40 Minutes)
DATA ANALYSIS

7) Experimental measurements of the rate of emission of radiant energy from the surface of a body shows that the rate of emission of radiant energy per unit area, E, varies with absolute temperature, T, as follows;

$$E = AT^n \text{ , where A and n are constants.}$$

The table below gives data for the rate of emission of radiant energy per unit area as a function of temperature in °C.

E/ 10^8 W m^{-2}	0/ °C
0.27	4456
0.85	5950
1.40	6798
2.00	7419
2.60	7917
3.20	8339
3.75	8706
4.28	9033
4.90	9329
5.50	9600

a) Plot a suitable graph that can be used to determine the values of A and n. **(11 marks)**
b) Use your graph to determine the values of A and n. **(7 marks)**
c) State the physical significance of A. **(2 marks)**

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SECTION III (50 Minutes)
OPTIONS
Answer any TWO questions from the four options.

OPTION 1: ENERGY RESOURCES AND ENVIRONMENTAL PHYSICS

8) a) Ethanol is a carbon neutral biofuel and a direct alternative energy source to petrol used in vehicles.
   i) Define the underlined phrases.
   ii) Why is ethanol described as a carbon neutral fuel?
   iii) Ethanol is capable of releasing 30 MJ of energy per kilogram. A vehicle moving at a steady speed of 25 m s^{-1} uses this fuel at a rate of 12 km per kilogram. Estimate the input power from the fuel to the engine. **(6 marks)**

   b) Energy can be obtained by splitting Uranium atoms in nuclear power stations and harnessing the ocean's tides.
   i) State one advantage to our environment, common to both ways of harnessing energy.
   ii) Identify one big environmental problem each, resulting from both ways of harnessing energy.
   iii) A tidal power station is operating at an efficiency of 40 %. It traps sea water of density 1050 kg m^{-3} over a level area of 15 m^2 when high tide is 3.0 m above low tide. Estimate the average power from the tidal power station. **(6 marks)**

   c) i) What is meant by space weather?
   ii) State one significant impact of space weather on earth and explain how it can be minimized. **(3 marks)**

OPTION 2: COMMUNICATION

9) a) A digital communication link between two towns 250 km apart is to be provided using an optic fibre.
   i) What is meant by a digital communication link?
   ii) State two changes likely to occur in a digital signal transmitted along an optic fibre.
   iii) Suggest why infra-red radiation is preferable over ultraviolet radiation for digital communication using optic fibres between the two towns. **(5 marks)**

   b) A large hill separates a householder from a communication mast or tower that transmits both radio and television programmes. A local radio station (Rnews) has a carrier wave frequency of 200 kHz and television station (TVnews) has a carrier wave frequency of 600 MHz. The householder can receive Rnews but cannot receive TVnews.
   i) Explain how the householder can receive Rnews even though the mast is on the other side of the hill and why he cannot receive TVnews.
   ii) Rnews broadcasts its signal using Amplitude Modulation (AM). If the highest frequency component in the audio signal is 5 kHz, calculate the bandwidth required for Rnews transmission.
   iii) Sketch a labelled frequency spectrum for the amplitude modulated radio waves. **(6 marks)**

   c) A mobile telephone handset is in its simplest form a radio transmitter and receiver.
   i) Draw a block diagram of the transmission section of a mobile phone handset.
   ii) Write down the full meaning of; GSM and PSTN in mobile phone technology. **(4 marks)**

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OPTION 3: ELECTRONICS

10) a) Figure 8 shows a circuit consisting of a Zener diode of voltage rating 50 V used to regulate the voltage across a fridge of resistance 7.5 kΩ.

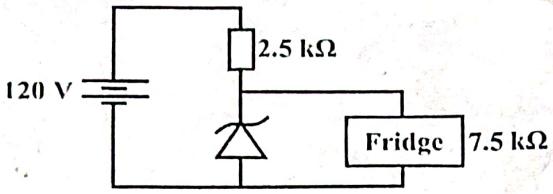


Figure 8

i) Show that the Zener diode is in the ON state.
ii) Why is a Zener diode used in voltage regulation?
iii) Explain why the Zener diode is used as a voltage regulator. **(4 marks)**

b) Figure 9 shows a BJT npn transistor circuit whose doping level at the base (B) is increased slightly.

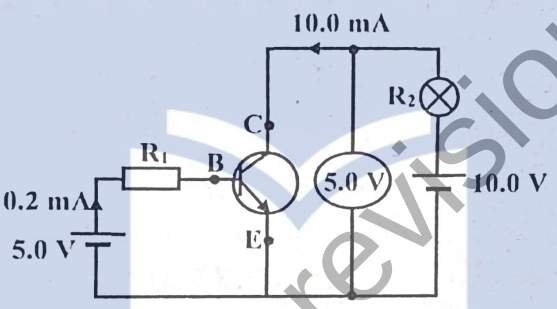


Figure 9

i) Define a transistor and state its mode of connection in the circuit.
ii) Neglecting the base – emitter voltage, determine the resistance of R_1 and R_2.
iii) If by subsequently altering R_1 only, V_{CE} were adjusted to 2.5 V, what would then be the value of the collector current? **(8 marks)**

c) Figure 10 is a block diagram of an electronic circuit.

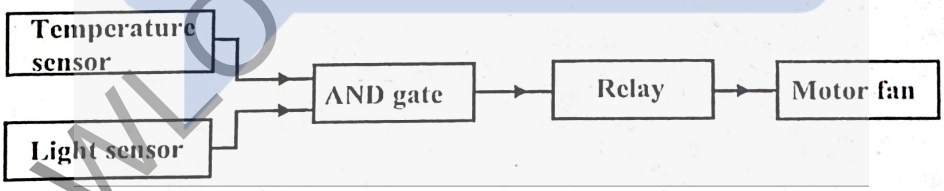


Figure 10

i) The temperature sensor is ON (1) when it is hot and OFF (0) when it is cold. The light sensor is ON (1) in daylight and OFF (0) in darkness. Explain what happens in each part of the circuit when it is both hot and in daylight.
ii) Describe a practical use for the circuit. **(3 marks)**

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OPTION 4: MEDICAL PHYSICS

11) a) The human heart beat rate of an athlete averages 180 beats per minute and pumping 90 ml of blood per beat during intense training for Mount Cameroon race 2025.

i) Draw a diagram showing the basic structure of the human heart acting as a double pump.

ii) Determine the total volume of blood pumped during his 2 hours training session.

(5 marks)

b) X-rays and MRI are all used in medical diagnosis.

i) State one situation in which each of these techniques is preferred, giving a reason for your choice.

ii) Explain why a barium meal might be given to a patient who is about to undergo an X-ray scan of the gastrointestinal tract. **(5 marks)**

c) Iodine-131 is a radioisotope used as a tracer to monitor the function of an internal organ of a patient.

i) What is meant by a radioactive tracer in medical diagnosis?

ii) Name the radiation emitted by the radioisotope for it to be detected by a detector outside the patient's body. Give a reason for your answer.

iii) State with a reason whether the radioactive tracer should have a short half-life or a long half-life. **(5 marks)**

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